

BARANOVICHI, Belarus

WMO: 269410

Lat: 53.1319N

Lon: 25.9719E

Elev: 635

StdP: 14.36

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.8	-8.2	3.6	-1.0	-2.2	5.0	3.7	21.0	31.9	18.7	30.5	4.5	30	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	83.8	67.4	80.8	66.1	77.9	64.5	69.9	79.7	68.2	77.2	66.5	74.9	7.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	99.6	74.2	64.8	94.2	72.7	63.2	88.9	70.9	34.2	79.8	32.8	77.1	31.5	75.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.8	13.9	-11.4	87.6	5.5	2.6	-15.3	89.4	-18.6	90.9	-21.7	92.3	-25.7	94.2		
(5)			-11.7	73.3	5.5	2.2	-15.6	74.9	-18.9	76.2	-21.9	77.5	-25.9	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	24.4	26.4	34.3	46.9	56.1	61.7	66.1	64.3	55.0	44.9	35.9	27.9		
	DBStd	16.71	10.71	10.37	8.37	7.86	7.59	6.14	5.65	5.55	6.14	7.34	8.23	9.85		
	HDD50	3446	793	662	489	149	28	0	0	0	23	191	425	686		
	HDD65	7350	1258	1082	953	544	291	137	55	81	303	622	873	1151		
	CDD50	1777	0	0	2	56	216	351	500	442	174	34	2	0		
	CDD65	203	0	0	0	1	14	37	90	58	3	0	0	0		
	CDH74	1623	0	0	0	11	134	287	724	437	30	0	0	0		
	CDH80	335	0	0	0	1	19	42	175	97	1	0	0	0		
(14) Wind	WSAvg	6.9	7.6	7.6	7.7	7.0	6.8	6.4	5.8	5.6	6.1	7.0	7.3	7.5		
(15) Precipitation	PrecAvg	24.0	1.3	1.1	1.4	1.6	2.2	3.1	3.2	2.7	2.1	1.9	1.8	1.6		
	PrecMax	30.9	3.4	2.6	3.0	4.1	4.5	5.7	8.0	8.3	6.9	5.6	3.6	3.2		
	PrecMin	17.3	0.4	0.2	0.2	0.5	0.8	1.2	0.6	0.3	0.4	0.1	0.5	0.5		
	PrecStd	3.8	0.7	0.6	0.6	0.8	1.0	1.2	2.0	1.6	1.3	1.3	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	47.1	59.2	75.2	82.7	84.6	88.6	87.6	77.9	68.5	55.9	47.0		
		MCWB	42.4	42.4	48.3	58.8	64.6	67.0	69.2	69.7	62.1	59.8	51.1	44.7		
	2%	DB	40.9	43.1	54.0	69.0	77.7	80.9	84.9	82.4	73.1	62.7	51.8	43.8		
		MCWB	38.9	40.2	45.0	54.0	62.8	66.2	67.9	67.5	60.6	55.8	49.3	42.1		
	5%	DB	38.6	40.4	49.7	64.6	73.7	77.5	81.3	78.8	69.1	58.5	48.7	41.3		
		MCWB	37.1	38.2	42.7	52.1	60.3	63.9	67.0	65.0	58.9	53.8	46.6	40.0		
	10%	DB	36.2	37.9	45.6	60.4	70.0	74.1	78.1	75.3	65.9	55.4	46.4	38.6		
		MCWB	35.0	36.2	40.7	49.7	57.8	62.7	65.7	63.5	57.4	52.1	44.7	37.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	43.8	49.6	59.7	68.3	70.1	73.2	71.4	64.6	60.3	52.1	45.3		
		MCDB	44.3	46.4	57.1	73.8	79.8	80.1	82.7	83.3	73.8	66.0	54.6	46.3		
	2%	WB	39.3	40.8	46.2	55.5	64.5	67.9	70.6	69.1	62.0	57.1	49.3	42.4		
		MCDB	40.5	42.6	52.1	65.3	74.1	77.4	80.6	79.1	70.0	61.4	51.5	43.4		
	5%	WB	37.2	38.4	43.5	53.1	61.9	66.0	68.7	67.5	60.3	54.6	47.0	40.1		
		MCDB	38.5	40.0	48.4	61.7	71.2	74.8	77.6	76.0	66.8	57.4	48.5	41.3		
	10%	WB	35.0	36.2	41.2	50.9	59.4	63.9	66.9	65.3	58.6	52.1	44.8	37.5		
		MCDB	36.1	37.8	45.2	58.6	67.9	71.5	75.4	72.1	64.4	55.4	46.1	38.5		
(35) Mean Daily Temperature Range	MDBR		7.3	8.6	11.4	15.7	17.4	16.2	17.0	17.0	15.5	11.3	7.0	6.7		
	5% DB	MCDBR	6.7	9.1	17.7	22.3	22.6	21.1	21.7	22.0	21.4	16.1	10.0	7.6		
		MCWBR	6.4	7.2	10.8	10.7	10.5	9.4	8.6	8.8	10.3	7.9	6.8			
	5% WB	MCDBR	6.6	8.8	15.7	18.9	19.9	18.0	18.7	17.5	17.6	14.0	9.0	7.3		
		MCWBR	6.4	7.3	10.5	10.1	10.2	9.1	8.6	8.1	9.9	9.8	7.5	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.330	0.364	0.413	0.403	0.391	0.424	0.420	0.383	0.354	0.337	0.297		
	taud		2.186	2.140	2.186	2.178	2.275	2.335	2.255	2.264	2.328	2.387	2.286	2.227		
	Ebn at Noon		210	238	256	258	268	273	260	254	249	232	193	186		
	Edh at Noon		21	30	36	42	40	38	40	38	31	24	20	17		
(44) All-Sky Solar Radiation	RadAvg		224	469	880	1307	1668	1784	1674	1452	990	520	198	140		
	RadStd		38	85	132	156	162	132	161	124	116	103	21	33		

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	+2.12	+2.65	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.79	N/A	N/A	N/A	+1.39	N/A	N/A	+162	N/A

Nomenclature: See separate page